

The Impact of Minimum Wage Increases on Evictions: Evidence from Rural and Urban Areas

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Abstract:

Housing instability and eviction are huge issues in the United States, and low-income renters are particularly vulnerable. The federal minimum wage has been set at \$7.25 since 2009, leaving many workers below the poverty line. Many states and counties have raised their minimum wages in an effort to improve affordability. Prior studies have found strong negative relationships between wage levels and housing instability; however, heterogeneity between rural and urban communities has not been explored. This study aims to examine how minimum wage increases affect eviction rates and if the response differs based on population density. Using data from the Eviction Lab, IPUMS ACS, US Census Bureau, Vaghul and Zipperer's minimum wage database, and the Bureau of Labor Statistics, we implement a series of two-stage difference in difference models using the methodology of Gardner (2021) to examine this relationship between 2010 and 2018. Despite many limitations regarding our data, our findings align with existing literature suggesting that minimum wage increases may reduce eviction rates. We find no significant relationship in rural counties (Estimate = -0.339, $p = 0.523$); however, we see a significant negative relationship in urban counties. (Estimate = -1.598, $p < 0.01$) Our work aligns with existing literature's findings of a negative relationship between minimum wage and eviction rates. The presence of a significant relationship only in urban counties suggests that policies aimed at reducing housing instability should be targeted at urban areas rather than at the federal level.

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Motivation/Background

Housing instability is a huge issue in the United States, and eviction, or risk of eviction, is a severe threat to many renters, particularly those working low-income jobs. It is estimated that in 2018, 7.8 evictions were filed per 100 renting households in the US, and landlords file around 3.6 million eviction cases every year. (The Eviction Lab, 2025) Furthermore, eviction causes cascading harms, leading to potential job loss, family disruption, and health decline.

At the same time, minimum wage laws are one of the most common policies aimed at helping reduce rent burden and housing instability. Currently, the federal minimum wage is \$7.25/hr, where it has been since July of 2009, and is officially considered a poverty wage, meaning a full-time worker earning minimum wage is living below the poverty threshold. As of 2022, this wage was worth almost 30% less than it was when originally implemented. (Cooper et al., 2022) Because of this, many states and local governments have raised the minimum wage to much higher than the federally mandated value. States like Washington and California have their minimum wages set at \$16.66/hr and \$16.50, respectively, and many local governments set the wage floor even higher than their state's requirements. (NCSL, 2025)

These two factors can create a feedback loop. As of July 2023, the National Low Income Housing Coalition reported that there are no states where someone working full-time at minimum wage would be able to afford a two-bedroom apartment. (Sarver, 2023) Workers earning minimum wage are no longer able to afford housing, which can lead to housing instability. In turn, housing instability presents several barriers to getting or retaining a job. Desmond & Gershenson (2016) show that eviction often leads to unemployment, and job loss also makes eviction more likely. In fact, 53% of adults in homeless shelters and 40% of people sleeping on the streets were employed in the same year they experienced homelessness. (Meyer et al., 2021). This highlights the disparity between current wage floors and costs of living, and the loop will continue until someone is able to secure a position that covers housing costs.

This study aims to answer the question: *How do changes in minimum wage laws affect eviction rates, and do these effects differ between rural and urban areas?* Specifically, we will examine whether increases in state or local minimum wages reduce (or increase) eviction filings, and whether these effects are stronger in rural versus urban areas.

One potential solution to address this feedback loop would be to raise the federal minimum wage to a livable wage. This would allow workers earning the wage floor to be able to at least afford housing, which would in turn increase their ability to remain employed. A key study in the minimum wage and eviction field utilizes a differences-in-differences model examining the effect of minimum wage increases on rent payment patterns, and finds that renters in states that increased the minimum wage had a significant reduction in monthly defaults, with 10.6% fewer

monthly defaults compared with renters in states without an increase. (Agarwal et al., 2022) These findings are corroborated in the literature across the field of minimum wage and housing instability.

A similar study employs a difference-in-difference model to estimate the effect of minimum wage on eviction rates, using five case studies of states with and without a minimum wage increase. They conclude that a 1 percentage point increase in the percentage deviation of a state's minimum wage from the federal minimum wage is correlated with a 1.92 percentage point decrease in eviction rates. (Zhang, 2020) In a similar model, the relationship between eviction rates and minimum wage is examined; however, only counties on either side of a state border are compared to control for unobserved spatial heterogeneity. (Dunphy, 2020) Together, these findings provide strong evidence for the relationship between minimum wage and housing instability.

While fairly extensive literature highlights connections between wage floors and eviction rates, policy changes are experienced differently across rural and urban areas due to differences in wages, rent burdens, cost of living, and access to social services. A federal minimum wage increase would produce very different effects on housing stability depending on the population density of that location. In 2021, it was estimated that a \$15/hr federal minimum wage would impact 42% of rural workers vs. 31% of urban workers (Chen et al., 2021), meaning rural workers are already more likely to be making only minimum wage. Since that time, several counties and cities have raised their minimum wage; however, these increases tend to be targeted more heavily towards urban areas, which could further widen the gap. Understanding potential heterogeneity between rural and urban areas is valuable in aiding decision-making around minimum wage laws. By further examining this, policymakers can work to more effectively reduce housing insecurity. I aim to fill this gap by exploring not only the connection between increases in minimum wage and eviction rates, but also how this connection may differ across rural vs. urban areas. In doing so, I seek to provide further evidence on the intersection between labor market policy and housing stability, and to highlight the role of geographic context on household outcomes.

Methods

Data Source

This project utilizes several data sources. Data on eviction filings comes from the Eviction Lab at Princeton University, a research organization with a database compiling the largest accumulation of US eviction court records ever compiled. The Eviction Lab provides county-level panel data from 2000-2018 on eviction filing rates, threatened rates, and judgment rates. Household demographics data comes from IPUMS ACS, which provides annual data from the American Community Survey on population and housing. Minimum wage data is from Vaghul and Zipperer (2022), who provide historical state and substrate minimum wages at various frequencies. Rural-urban classification is calculated using the US Census Bureau rural-urban data. Counties with more than 50% population living in urban areas are classified as

urban, and the others are classified as rural. We use historical county-level unemployment data from the US Bureau of Labor Statistics.

The full dataset links these sources from 2010 to 2018 at the county level. We aggregate the American Community Survey data to the household level to reflect units potentially at risk of eviction.

Variables of Interest

Our outcome variable of interest is threatened rates, which are calculated by the Eviction Lab as the proportion of households that ever received a filing in a given year. We use this instead of the filing rate because the same household can receive multiple filings in a year, which could inflate the data. The threatened rate can give us a good idea of how many households are not able to afford their rent.

Control variables include covariates that could be potentially correlated with both eviction rates and minimum wage changes in a county. Our controls include:

- Share of properties owned rather than rented (ownership_share)
- Share of college-educated residents (share_college)
- Share of employed females (share_female_employed)
- Share of immigrants (share_immigrants)
- Unemployment rate (unemp_rate)

By including these, we are adjusting for potentially confounding factors that could explain some of the variation in eviction threatened rates and impact our results.

We also incorporate fixed effects and clustering into our models. County fixed effects intend to control for unobserved county characteristics that remain constant over time, while year fixed effects intend to control for any national shocks or trends over time that occurred in all counties. Our standard errors are clustered at the county level, which accounts for heteroskedasticity among counties.

For this analysis, we restrict our sample to counties that experience a minimum wage increase at some point during the study period. This means that all counties in the dataset receive treatment at some point. Because increases occur in a staggered manner, the control group is defined as counties that have not yet experienced the minimum wage increase, and the treatment group is defined as counties in periods after their minimum wage increase.

Event Study

We run an event study using a fixed-effects OLS regression to estimate the average difference in threatened rates in each event-time period relative to one year before treatment. This approach compares each county to itself at different relative years to treatment, which shows the trends that counties were on relative to their pre-treatment reference period. We expect to

see insignificant coefficients across the pre-treatment time, as this would show that there are no substantial baseline differences or trends in our counties, and changes we see in the post-treatment can be attributed to the minimum wage increase.

Our event study specification is:

```
event_model <- feols(
  threatened_rate ~ i(event_time, treated, ref = -1) + ownership_share +
  share_college + share_female_employed + share_immigrants +
  unemp_rate | fips + year,
  data = df)
```

This highlights any remaining variability that is not being controlled for by our covariates.

Difference-in Difference

We aim to run a difference-in-difference (DiD) model investigating the impact of minimum wage increases on eviction rates. Treatment is defined as the first increase in minimum wage in the county occurring during the period (2010-2018).

The DiD compares changes in eviction threatened rates within counties over time. It contrasts the treated counties' threatened rates before and after a change in minimum wage relative to the threatened rates in the same relative time period in the control group.

We utilize the did2s two-state estimator following Gardner (2021). This first regresses the threatened rate on our controls and fixed effects before regressing those residuals on our minimum wage treatment indicator. This allows us to obtain a cleaner estimate of how the treatment of a minimum wage increase affects threatened rates, thereby decreasing the concern of heteroskedasticity. The treatment effect can be interpreted as the average change in threatened rates resulting from an increase in the minimum wage. These models are run separately for rural and urban counties to investigate potential heterogeneity by population density.

We first run a static model for each population density classification (rural & urban), with:

$$Y_{it} = \alpha_0 + \alpha_1 \text{ownership_share}_{it} + \alpha_2 \text{share_college}_{it} + \alpha_3 \text{share_female_employed}_{it} \\ + \alpha_4 \text{share_immigrant}_{it} + \alpha_5 \text{unemp_rate}_{it} + \gamma_i + \delta_t + \varepsilon_{it}$$

$$\tilde{Y}_{it} = \beta_0 + \beta_1 \cdot D_{it} + u_{it}$$

Where:

- **Eviction Threatened Rates (Y_{it})** is the outcome

- **Control Variables** include previously specified county-level characteristics
- **County Fixed Effects** (γ_i) control for time-invariant county characteristics
- **Year Fixed Effects** (δ_t) control for events that affected all counties in the same year
- **Error Term** (ε) explains any remaining variability after regressing eviction rates on our controls and fixed effects

And:

- $(\hat{Y}_{it})$ is the residualized outcome
- **Treated** (D_{it}) is equal to 1 for counties that experienced a minimum wage in the years after the policy change occurred, and 0 otherwise

With this specification, we aim to isolate the effect of a minimum wage change on eviction-threatened rates.

Dynamic DiD

Because minimum wage changes were implemented in different years for different counties, we implement a dynamic difference-in-difference study to look at the treatment effects over time before and after the treatment. This provides insight into how the effects may evolve or persist over time.

To do this, we construct a time variable that indicates years relative to the treatment. Negative values represent years before the minimum wage change, and positive values represent years after. We estimate the dynamic effects by including indicators for each relative time period in the second stage of the did2s model. The reference period is set to one year before treatment, as this allows us to assess pre-treatment trends. Again, separate models are run for rural and urban counties, and we utilize the two-step estimator following Gardner (2021).

We hypothesize that rural counties will experience greater effects from a minimum wage increase than their urban counterparts, as rural counties tend to have lower housing costs and overall lower costs of living. Additionally, prior research has found that people living in rural areas are more likely to be making minimum wage, meaning they would more likely be affected by an increase (Chen et al., 2021).

The specification for these models follows the same structure as before, and is as follows:

$$Y_{it} = \alpha_0 + \alpha_1 \text{ownership_share}_{it} + \alpha_2 \text{share_college}_{it} + \alpha_3 \text{share_female_employed}_{it} \\ + \alpha_4 \text{share_immigrant}_{it} + \alpha_5 \text{unemp_rate}_{it} + \gamma_i + \delta_t + \varepsilon_{it}$$

$$\tilde{Y}_{it} = \beta_0 + \sum_{k \neq -1} \beta_k \cdot \mathbf{1}(\text{rel_year} = k)_{it} + u_{it}$$

Comparing these results will enable us to determine whether rural or urban counties are affected differently or to varying degrees by a minimum wage increase.

Results

Data

Our sample includes data from 2010 to 2018. This period lies between the Great Recession and the onset of the coronavirus pandemic, both of which would have had significant effects on eviction rates and therefore would have a possibility of muddying the effects of our analyses. Summary statistics can be found below in Table 1.

Characteristic (Rural)	Pre-Treatment	Post-Treatment
threatened_rate	5.2 (3.4, 10.1)	5.1 (3.9, 7.4)
ownership_share	0.72 (0.70, 0.77)	0.71 (0.68, 0.75)
share_college	0.23 (0.18, 0.31)	0.22 (0.19, 0.30)
share_female_employed	0.65 (0.63, 0.69)	0.67 (0.64, 0.69)
share_immigrant	0.04 (0.02, 0.09)	0.03 (0.02, 0.07)
unemp_rate	8.10 (7.20, 10.10)	5.30 (4.50, 6.60)

Characteristic (Urban)	Pre-Treatment	Post-Treatment
threatened_rate	5.2 (3.5, 8.4)	5.5 (3.5, 7.8)
ownership_share	0.64 (0.59, 0.72)	0.63 (0.58, 0.69)
share_college	0.34 (0.28, 0.40)	0.32 (0.28, 0.39)
share_female_employed	0.67 (0.63, 0.70)	0.67 (0.65, 0.70)
share_immigrant	0.12 (0.08, 0.19)	0.13 (0.08, 0.18)
unemp_rate	7.70 (6.00, 9.90)	5.40 (4.40, 7.20)

Table 1: Summary statistics and progression trends of dependent and control variables by rural (top) and urban (bottom) classification.

Event Study

The event study highlights significant coefficients in some pre-treatment periods for both rural and urban counties, indicating that counties' pre-treatment trajectories have baseline differences, raising concerns about a violation of the parallel trends assumption.

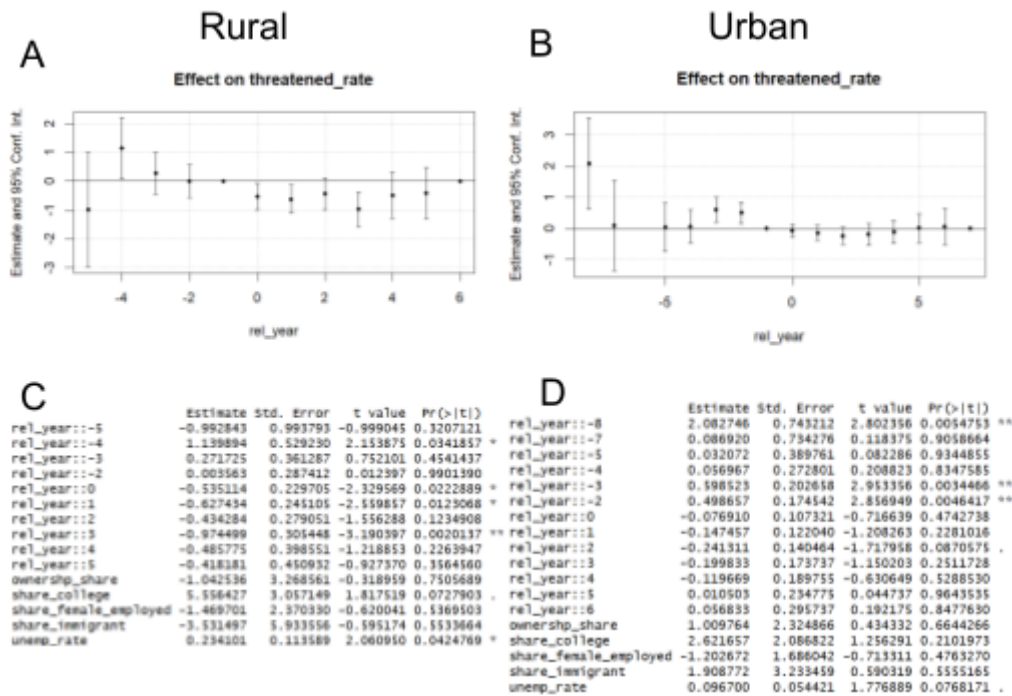


Figure 2: Fixed-effects OLS regression event study. A and B show the plotted estimates and standard errors for each year relative to treatment for rural (left) and urban (right) counties, while C and D show the full regression output for each model.

Difference-in-Difference

Our static DiD models found negative associations between minimum wage changes and threatened rates. Urban counties experienced a significant relationship (Estimate = -1.598, $p < 0.01$), while rural counties showed an insignificant, but negative, association (Estimate = -0.339, $p=0.523$). This would indicate the relationship is much stronger in urban counties than in rural ones, contrary to our original hypothesis. However, given the violation of the parallel trends assumption, these results should be interpreted cautiously, and causal inference cannot be made. County and year fixed effects were included in the model in an attempt to adjust for observed differences in the data, but they are not able to fully resolve the heterogeneity.

Dynamic Difference-in-Differences

Estimates from the dynamic DiD models echo the event study findings: significant pre-treatment effects exist in both rural and urban models. This suggests that counties are on different trajectories before experiencing treatment, and means we are unable to disentangle the true relationship between minimum wages and eviction rates. The event study and coefficients are found in Figure 3.

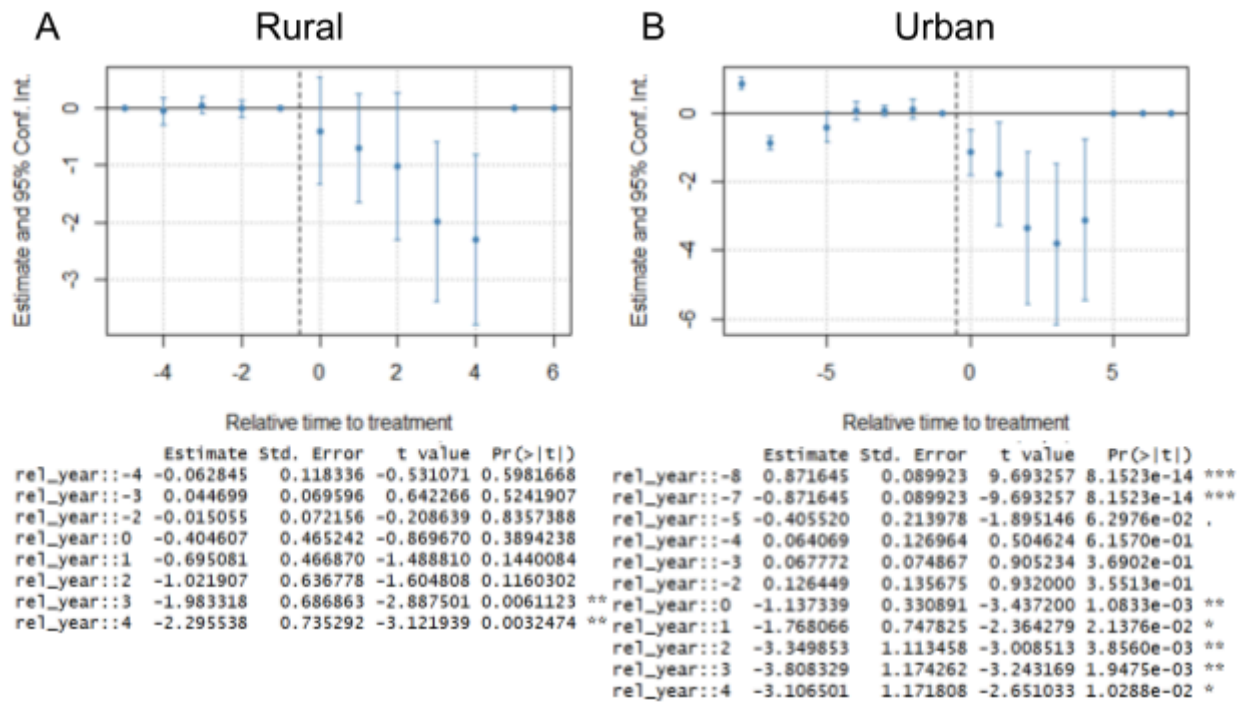


Figure 3: The event study model for the DiD model highlights the relationship between minimum wage increases and eviction threatened rates between the treated and untreated groups before and after the implementation of a minimum wage increase.

We expect to see no significant estimates before treatment, as this indicates that our control and treatment groups are similar and can be compared. This does seem to hold true for the rural cohort; however, we see significant coefficients in the periods 7 and 8 years before treatment in our urban group. This undermines any claims of causal impact, as the true effects are unable to be disentangled from group differences.

For the rural cohort, there are significant negative coefficients at 3 and 4 years post-treatment. Despite seeing no overall significant effect from our static model, this highlights that there may be a lagged response from when the minimum wage is changed to when rural counties see an effect. This may be because an increase does not lead to a sudden increase in income; rather, it is earned over time, and it may take a few years for people to start being able to afford their housing. Unfortunately, our data does not contain enough time to see if this relationship continues beyond 4 years or if adjustments in housing costs respond and bring the eviction rate back up.

For the urban cohort, we see significant negative coefficients for all observed post-treatment periods (0-4 years). This aligns with our highly significant result from the static model and further demonstrates the negative response. For urban counties, threatened rates reach their lowest around 3 years post minimum wage change, as the coefficient for year 4 is slightly smaller in

magnitude. However, we would need extended data to more fully see the prolonged evolution or persistence of the effect.

Discussion

Although eviction levels greatly dropped due to a moratorium put into place over the coronavirus pandemic, evictions in many areas have now returned to or exceeded pre-pandemic levels (Eviction Lab, 2025). Currently, several factors are understood to drive housing instability, including the stagnant federal minimum wage and housing costs that have doubled since the last federal minimum wage increase (FRED). Working to fully understand how each factor impacts eviction can allow for effective policy change (Moini & Elder, 2024).

To put this data into perspective, as of 2023, 1.1% of hourly workers made only the minimum wage, and many more earned near-minimum wages (US Bureau of Labor Statistics, 2024). This leaves a lot of people working full-time and still living below the poverty line.

The link between wage floors and evictions is strong. As mentioned earlier, earning insufficient income leads to rent burden, which can then lead to housing instability, which leads to employment instability, which further leads to eviction risk.

Results from our DiD models tentatively suggest that overall higher minimum wages are correlated with reduced threatened rates. This implies that the increased income seems to outweigh potential raises in rent or crowding out by landlords, and the income effect dominates the price effect. However, this being said, it is important to underscore the violation of the parallel trends assumption, meaning we are not able to claim that this relationship is caused by the minimum wage increases.

Our investigation into the heterogeneous effects between rural and urban counties found that urban counties had a strong, negative relationship between minimum wage and eviction threatened rates, while there was no significant relationship found in rural counties. This suggests that minimum wage increases may be better targeted at urban areas than nationwide, as this seems to be where the effect is largest. This may align with the theory of Andrews (2022), who found that higher minimum wages could actually have an adverse effect on rural counties by harming growth and employment in rural areas. This is due to the existence of less income inequality and the overall lower cost of living in those zones. His research suggests that, regarding housing instability, rural areas may not benefit from a minimum wage increase. However, we did still see a negative relationship, suggesting that there are not necessarily adverse effects. Furthermore, the dynamic DiD verifies this negative trend.

This study has several limitations that should be acknowledged. Substantial baseline differences seen in the pre-treatment period indicate that, despite using the yet-to-be-treated counties as our control, there may be other effects that are not controlled for coming into play. Furthermore, because of the violations of the parallel trends assumption, estimates cannot be interpreted in a

causal manner. This is likely why previous studies have exclusively utilized a matching approach when examining this relationship.

Additionally, our data itself is limited. Eviction lab data is not comprehensive of all counties, which limits our geographic scope, and is only measured at a yearly level, meaning seasonal variations are not captured. Data from the American Community Survey also does not span every county, meaning our sample size is limited further. Although we did use adjusted models, we are not able to rule out other confounders, or time-varying confounders such as economic factors or other policy interventions.

As this study does not yield interpretable results to determine causation, there are many methodological improvements that should be considered for future work. These include estimators proposed by Sallaway and Sant'Anna (2020), which could be used as a robustness check or future extensions. Looking into a synthetic control or matching DiD could also potentially help with the violation of the parallel trends assumption.

Further improvements, which include using more comprehensive data with coverage of more counties and/or more frequent time intervals, would increase our sample size and lead to a more robust model. Including more covariates may also help us control for more unexplained variation and further disentangle the true effects.

Future steps for this research could include using a matching approach or synthetic control to amend our violation of the parallel trends assumption and allow for interpretable, causal results. Additionally, considering the amount of deviation from the federal minimum wage may be able to more effectively outline policy implications for both rural and urban areas.

Although we are limited in causal inference, patterns seen in our data, as well as existing literature, suggest that wage increases are likely associated with lower eviction rates in urban counties. By understanding how wage floors are associated with threatened rates, policies can be better aimed at helping reduce housing instability. Particularly, policies can be specifically targeted toward urban areas rather than at the federal level to provide the most effective results.

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